

University of Windsor
Department of Mathematics and Statistics
Mathematical Foundations MATH 1020
Practice Questions for Midterm Test 2

1. Prove $\forall x, y, z \in \mathbb{Z}$, if x divides y and y divides z , then x divides z .
2. Prove $\forall x, y \in \mathbb{Z}$, if xy is even and y is odd, then $x + y$ is odd.
3. Prove $\forall x, y \in \mathbb{Z}$, if $x + y$ is odd, then $x - y$ is odd.
4. Prove $\forall x, y \in \mathbb{Z}$, if 3 divides xy and 3 divides $y - 1$, then 3 divides x .
5. Prove $\forall x \in \mathbb{Z}$, if 2 divides x and 5 divides x , then 10 divides x .
6. Prove $\forall a, x, y \in \mathbb{Z}$, if a divides $x + y$ and a divides $2x + y$, then a divides x and a divides y .
7. Using a proof by contradiction and the well-ordering property, prove $\forall x \in \mathbb{Z}$, if x is odd, then $\forall n \in \mathbb{N}$, x^n is odd.
8. Using a proof by contradiction and the well-ordering property, prove $\forall x \in \mathbb{R}$, if $2 < x$, then $\forall n \in \mathbb{N}$, $2^n < x^n$.
9. Prove $\forall x \in \mathbb{R}$, if $0 \leq x < 1$, then $\exists n \in \mathbb{N}$, $\frac{n-1}{n} \leq x < \frac{n}{n+1}$.
10. Using the principle of mathematical induction, prove $\forall x \in \mathbb{Z}$, if $\exists a \in \mathbb{Z}$, $x = 6a + 3$, then $\forall n \in \mathbb{N}$, $\exists b \in \mathbb{Z}$, $x^n = 6b + 3$.
11. Using the principle of mathematical induction, prove $\forall x \in \mathbb{Z}$, if 3 divides $x - 1$, then $\forall n \in \mathbb{N}$, 3 divides $x^n - 1$.
12. Using the principle of mathematical induction, prove $\forall a, x, y \in \mathbb{Z}$, if a divides $x - y$, then $\forall n \in \mathbb{N}$, a divides $x^n - y^n$.
13. Using the principle of mathematical induction, prove $\forall x \in \mathbb{R}$, if $0 < x < 1$, then $\forall n \in \mathbb{N}$, $x^n < 1$.
14. Using the principle of mathematical induction, prove $\forall n \in \mathbb{N}$, $\sum_{k=1}^n 4k + 3 = 2n^2 + 5n$.
15. Using the principle of mathematical induction, prove $\forall n \in \mathbb{N}$, $\sum_{k=1}^n \frac{1}{2^k} = 1 - \frac{1}{2^n}$.
16. Prove $\forall a, b \in \mathbb{R}$, if $a < b$, then $(-\infty, b) \setminus (-\infty, a) = [a, b)$.
17. Prove $(-\infty, 3] \setminus (1, 2) = (-\infty, 1] \cup [2, 3]$.
18. Prove $\langle 5 \rangle \cap \langle 4 \rangle = \langle 20 \rangle$.
19. Prove $\forall m, n \in \mathbb{Z}$, if m divides n , then $n + \langle m \rangle = \langle m \rangle$.
20. Prove $\langle 10 \rangle + \langle 15 \rangle = \langle 5 \rangle$.